GROUP WORK PROJECT #02
GROUP NUMBER: 6242

MScFE 600: FINANCIAL DATA

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Mohammed Khalid Siddiqui	United States	mohammedks713@gmail.com	
Olubukunola Anuoluwapo Ricketts	Nigeria	rbukkie@yahoo.com	
MD RASHEDUL ISLAM RASEL	BANGLADESH	mdrashedul207@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

Team member 1	**Mohammed Khalid Siddiqui**
Team member 2	**Olubukunola Anuoluwapo Ricketts**
Team member 3	**MD RASHEDUL ISLAM RASEL**

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.
Note: You may be required to provide proof of your outreach to non-contributing members upon request.

Table of Contents

0
Step 1: Magnifying Risks 2
Step 2. Frictional Related Challenges 4
Step 3. Identifying Additional Data 5
Step 4. Go get the data 7
Step 5. Perform exploratory data analysis 7
Step 6. Describe how the data can help to meet the challenge 10
References 12

Step 1: Magnifying Risks

Scenario	Leverage	Non-Linearities
Money at a fixed rate for an unsecured purchase (e.g. credit card) for an individual	1. Giving credit limits much higher than expected payable amounts of the part of the borrower can create a situation where the borrower can be levered significantly. Reckless spending on their part can cause them to go into debt with a high amount of interest-bearing debt they must service. This increases default risk. 2. The borrower may already have a significant amount of debt before the loan is given. This will further increase leverage and cause the loan to have a higher risk of default	1. Credit card debt has to compound interest. Overleveraged debt can, therefore, accumulate further debt on the part of the borrower in a non-linear fashion. Penalty payments and such can create an exponential spiral. 2. Due to default risk, the payoff is non-linear. The amount of the loan to be paid back will depend on the recovery rate in case of default.
Money at a floating rate for a secured purchase (e.g. home or automobile) for an individual	1. The amount borrowed is 80% of the underlying asset value, and the down payment is 20%. This creates a 5:1 leverage on the equity.	1. A mortgage is essentially an option on the house. If house value drops, a homeowner may default without recourse. Even in a solvent situation, variable mortgage rates can introduce a non-linear element to the financial gain.
Money at a fixed rate for a business for a construction loan	1. When financing construction projects, businesses may engage in high-leverage financing behaviors. Any hiccups in the project,	1. There are initial negative cash flows for the borrower and lender during the construction phase. The long-term expected

	such as delays or unexpected costs, can amplify leverage risks. 2. Part of the cost of financing the construction may already consist of significant debt prior to the loan being given, which will further increase the leverage.	payoff involves large but unexpected cash inflows.
Publicly traded Equity (e.g. common stock)	1. Investors can borrow funds to buy public equities, trading on margin. When the market experiences a downturn, this can exacerbate potential losses. 2. The borrower of the publicly traded equity is required to put 150% of the value of the shares borrowed in the margin account. This 150% includes 100% from short selling the borrowed shares, so the borrower is only putting up 50% of the value of the shares sold. This allows for leverage profit.	1. The combination of market movements, volatility, investor sentiment and other factors can create non-linear price movements for stocks. This excludes the non-linear impact of derivatives like options. 2. The payoff from short selling is non-linear. Gains are limited, while losses are unlimited.
Publicly traded bond (e.g. treasury bond, corporate bond)	1. Bonds can be used to collateralise further borrowing and also experience price shifts due to changing interest rates. Levered bond portfolios will experience nonlinear shifts in their value as a result.	1. The relationship between bond prices and interest rates in a volatile market is not linear.
An illiquid security – you choose	1. In private equity,	1. Illiquid securities can

the security	leveraged buyouts can amplify risk if the underlying asset underperforms. Due to the difficulty of selling the asset quickly due to its illiquid nature, private businesses or other illiquid securities may involve highly leveraged transactions.	experience significant non-linear movements in value. Specific events like IPOs or buyouts can introduce significant nonlinear changes in value.

Step 2. Frictional Related Challenges

Scenario	**Liquidity**	**Regulation**
Money at a fixed rate for an unsecured purchase (e.g. credit card) for an individual	1. Issuers of credit cards or similar lending products need to have short-term liquidity to cover obligations and cushion in case of borrower defaults during adverse market conditions.	1. The CARD Act and other regulations impose limits on transparency, predatory lending, and anti-discriminatory lending.
Money at a floating rate for a secured purchase (e.g. home or automobile) for an individual	1. Potential fluctuations in repayment rates and risks associated with refinancing or selling collateral require the lenders to manage liquidity.	1. Truth in Lending Act and Real Estate Settlement Procedures Act are two lenders that must adhere to the requirements regarding adjustable rate mortgages and disclosure requirements.
Money at a fixed rate for a business for a construction loan	1. Construction loans require significant liquidity. This is because of ongoing costs and management of the timing of loan payments and disbursements.	1. There are local, state, and federal laws & regulations regarding lending practices, environmental impact assessments, and construction permits. There are also banking regulations and capital

		requirements for business loans.
Publicly traded Equity (e.g. common stock)	1. There can be volatility in equity markets, especially for stocks with lower liquidity. This can impact trading strategies.	1. There are regulations pushed by the SEC and other bodies regarding financial reporting, insider trading, and market manipulation, among other behaviors.
Publicly traded bond (e.g. treasury bond, corporate bond)	1. Liquidity depends on the bond issuer, maturity, market conditions, and other factors. It may be hard to borrow some bonds because of market fragmentation.	1. There are regulations surrounding bond issuance and trading. Investor protections, disclosure, ratings, and regulatory oversight to prevent excessive borrowing and maintain market stability are important factors to consider.
An illiquid security – you choose the security	1. Illiquid securities are difficult to sell quickly. Finding a buyer can take a while and may be at a discount. Finding short-term value on an illiquid asset can be difficult.	1. There are government regulations one must comply with regarding private investments. There are disclosure requirements, investor qualification criteria, and anti-fraud regulations.

Step 3. Identifying Additional Data

Scenario	Data Type	Data Processing	Data Frequency	Data Class	Data Source	Data Variety
Money at a fixed rate for an unsecured purchase (e.g. credit card) for an	Credit	Categories	Quarterly	Credit	Consumer Credit Agencies	Observed Data

individual						
Money at a floating rate for a secured purchase (e.g. home or automobile) for an individual	Asset	Raw Prices	Monthly	Real Estate	Mortgage Lenders	Trade Data
Money at a fixed rate for a business for a constructio n loan	Asset	Raw Prices	Monthly	Real Estate	Constructio n Firms	Actual Data
Publicly traded Equity (e.g. common stock)	Asset	Raw Prices	Daily	Equity	Exchanges	Trade Data
Publicly traded bond (e.g. treasury bond, corporate bond)	Ratings	Levels	Daily	Credit	Rating Agencies	Quote Data
An illiquid security – you choose the security	Asset	Raw Prices	Daily	Small cap Equity	Exchanges	Trade Data

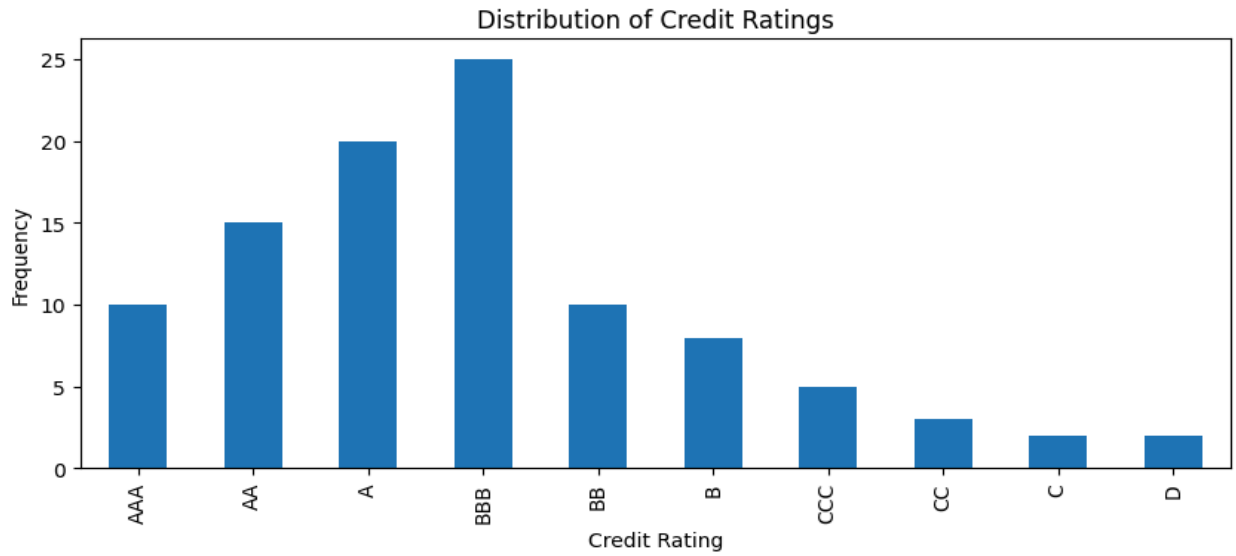
Step 4. Go get the data

(MD RASHEDUL ISLAM RASEL)

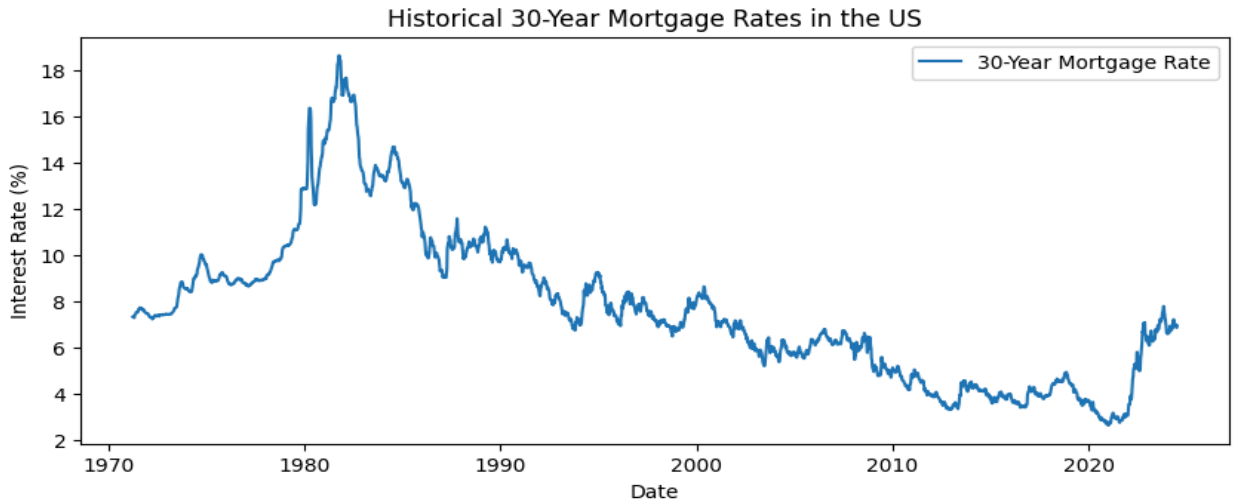
In order to maintain simplicity, we will use sample data from Yahoo Finance, the Federal Reserve Bank of St. Louis, and other publicly available sources.

Step 5. Perform exploratory data analysis

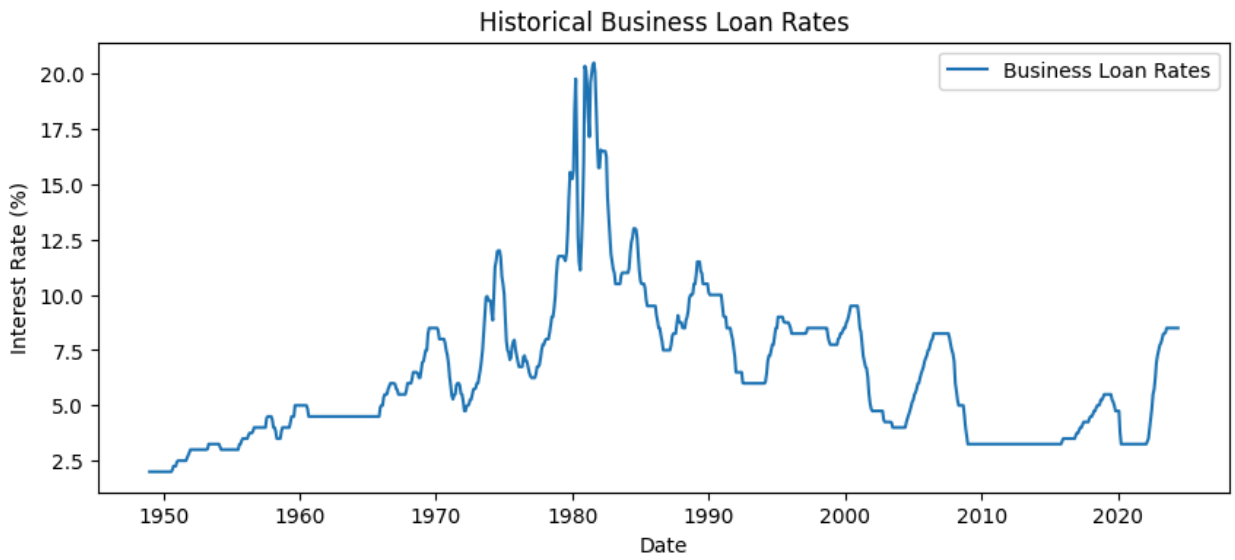
(MD RASHEDUL ISLAM RASEL)



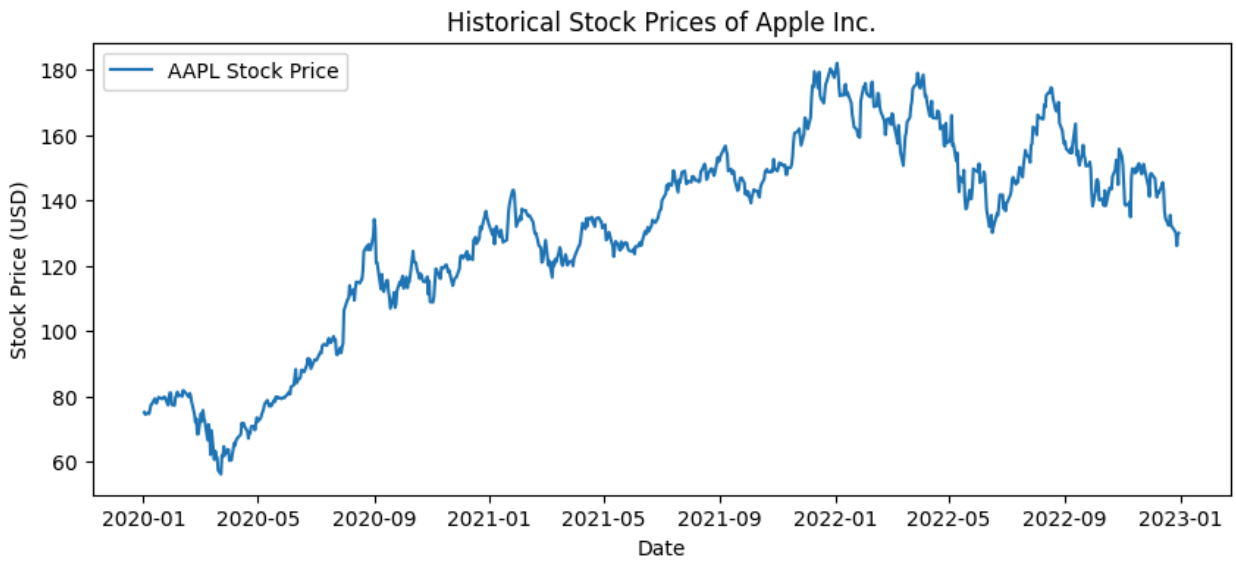
Insight: Credit ratings are concentrated in the mid to high range (BBB to AA), signifying diverse degrees of credit risk among borrowers.



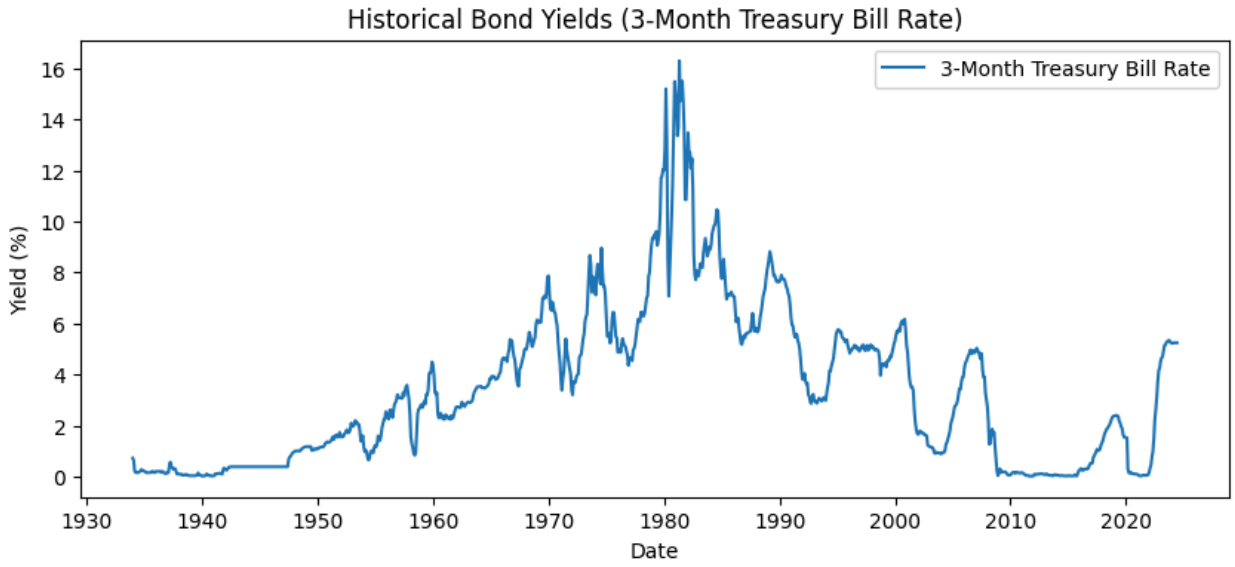
Insight: Fluctuations in US 30-year mortgage rates suggest potential volatility in borrowing costs for floating-rate mortgages.



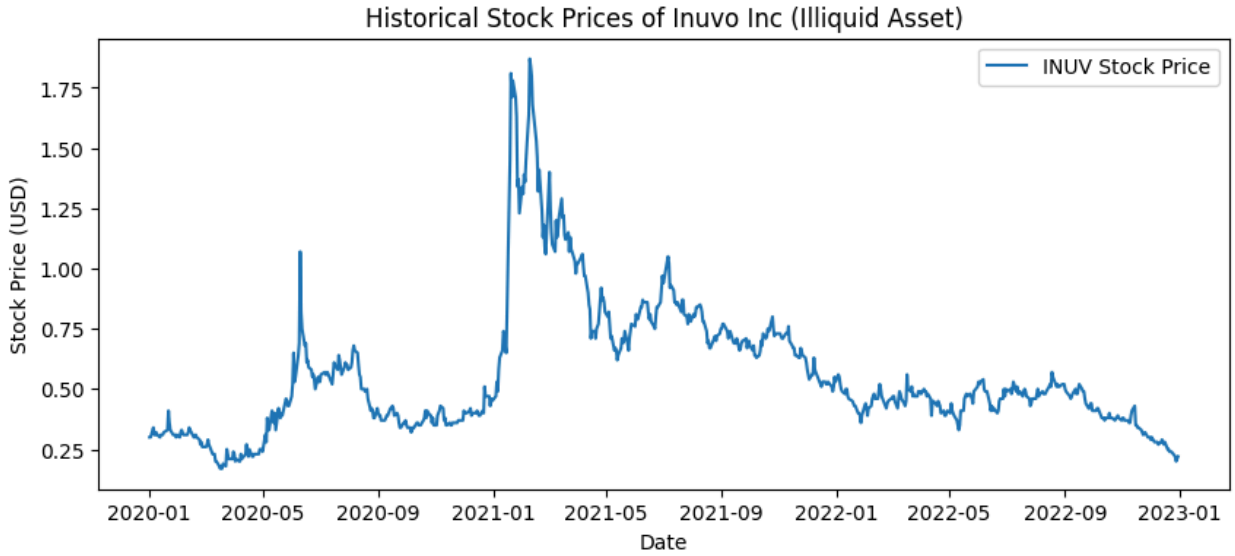
Insight: The variability in business loan rates reflects the changing cost of borrowing for businesses, impacting their financial planning and risk management.



Insight: The volatility of Apple Inc.'s stock price has displayed significant peaks and troughs, suggesting potential risks for equity investments.



Insight: Bond yields, as reflected in the 3-month Treasury bill rate, show the interest rate environment and the cost of borrowing in the short term, which can affect the attractiveness of bonds as investment vehicles.



Insight: The stock price of Inuvo Inc, a small-cap company, exhibits higher volatility and lower liquidity compared to large-cap stocks, indicating potential challenges in trading and valuation.

Step 6. Describe how the data can help to meet the challenge

(MD RASHEDUL ISLAM RASEL)

In this step, we'll examine how the data analysed in Group Work Project 1 (GWP1) and Group Work Project 2 (GWP2) can address financial challenges related to volatility, leverage, non-linearity, liquidity, and regulation.

Scenario 1: Fixed Rate for Unsecured Purchase (Credit Card for an Individual)

Data Used: Credit Scores

- Insights on Volatility: Credit scores provide insight into borrower creditworthiness and default risk. Analysing score distribution helps assess credit portfolio volatility.
- Insights on Leverage and Non-Linearity: Credit scores can reveal potential leverage, with lower scores disproportionately increasing default risk.

Scenario 2: Floating Rate for Secured Purchase (Home or Automobile for an Individual)

Data Used: Mortgage Rates

- Insights on Volatility: Mortgage rate fluctuations impact borrowing costs for secured loans, affecting borrower repayment ability.
- Insights on Leverage and Non-Linearity: Fluctuating rates affect borrower leverage and monthly payments.

Scenario 3: Fixed Rate for Business (Construction Loan)

Data Used: Business Loan Rates

- Insights on Volatility: Historical business loan rate trends impact borrowing costs and business operations.
- Insights on Leverage and Non-Linearity: Interest rate changes can significantly affect business leverage.

Scenario 4: Publicly Traded Equity (Apple Inc.)

Data Used: AAPL Stock Prices

- Insights on Volatility: Historical stock prices inform investors on equity market volatility and associated risks.
- Insights on Leverage and Non-Linearity: Equity investments can be leveraged, and non-linearity in equity returns is evident.

Scenario 5: Publicly Traded Bond (3-Month Treasury Bill Rate)

Data Used: Bond Yields

- Insights on Volatility: Bond yield data helps understand interest rate volatility, influencing bond investment risk.
- Insights on Leverage and Non-Linearity: Non-linearity during interest rate changes affect bond portfolio valuation and returns.

Scenario 6: Illiquid Security (Inuvo Inc.)

Data Used: INUV Stock Prices

Insights on Volatility: Historical stock prices indicate small-cap stock volatility.

Insights on Leverage and Non-Linearity: Small-cap stock price movements can be more drastic.

Key Observations

Volatility and Liquidity:

- Data from GWP1 and GWP2 enhance understanding of volatility in different financial scenarios. For instance, Apple Inc. and Inuvo Inc.'s stock prices exhibit varying volatility levels, highlighting risks associated with large-cap versus small-cap stocks.
- Inuvo Inc. data shows liquidity challenges, with low trading volumes indicating higher illiquidity.

Leverage and Non-Linearity:

- Mortgage and business loan rates highlight leverage challenges, impacting borrowing costs and ratios for individuals and businesses.
- Non-linearity can be observed in credit scores and stock prices, offering insight into financial market dynamics.

Regulation:

- Bond yield data offers regulatory insights, aiding the assessment of bond investment risk.

Additional Data:

- The VIX (volatility index) could further analyse market volatility's impact on different securities.

References

1. "Demystifying Private Equity Valuations | Russell Investments." Russellinvestments.com, russellinvestments.com/us/blog/demystifying-private-equity-valuations. Accessed 8 July 2024.
2. Nuveen. "Duration: A Measure of Bond Price Volatility." Nuveen, Nuveen, May 2024, www.nuveen.com/en-us/insights/municipal-bond-investing/understanding-duration#:~:text=Bond%20volatility%20refers%20to%20the. Accessed 8 July 2024.
3. "The Laws That Govern the Securities Industry | Investor.gov." Www.investor.gov, www.investor.gov/introduction-investing/investing-basics/role-sec/laws-govern-securities-industry#:~:text=Often%20referred%20to%20as%20the. Accessed 8 July 2024.
4. Federal Reserve Bank of St. Louis. (n.d.). *Welcome to fred, Federal Reserve Economic Data. your trusted source for economic data since 1991.* FRED. https://fred.stlouisfed.org/
5. *Large Bank consumer credit card balances: Current credit score: 50th percentile*. FRED. (2024b, April 15). https://fred.stlouisfed.org/series/RCCCBSCOREPCT50
6. *30-year fixed rate mortgage average in the United States*. FRED. (2024b, July 3). https://fred.stlouisfed.org/series/MORTGAGE30US
7. *Bank Prime Loan Rate*. FRED. (2024c, July 1). https://fred.stlouisfed.org/series/MPRIME
8. *3-month Treasury Bill Secondary Market Rate, discount basis*. FRED. (2024a, July 1). https://fred.stlouisfed.org/series/TB3MS